Multiple Linear Regression – Multi-Channel Marketing Analysis

Project Overview

This project focuses on predicting **Sales** using multiple marketing channels simultaneously. Unlike Simple Linear Regression, Multiple Linear Regression allows us to measure the impact of several independent variables while controlling for the effects of others.

The dataset contains:

Variable	Description
TV	TV advertising spend category
Radio	Radio advertising spend
Social Media	Social media advertising spend
Influencer	Influencer marketing category
Sales	Total sales generated (Target Variable)

Import Required Libraries

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm

from statsmodels.stats.outliers_influence import variance_inflation_factor
from scipy import stats
```

Load the Dataset

```
df = pd.read_csv('marketing_sales_data.csv')

df.head()
```

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

Initial Exploratory Data Analysis (EDA)

Dataset Shape

```
df.shape
```

```
(572, 5)
```

Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
```

```
---  ------        --------------  -----
 0   TV            572 non-null    object
 1   Radio         572 non-null    float64
 2   Social Media  572 non-null    float64
 3   Influencer    572 non-null    object
 4   Sales         572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

Missing Values

```
df.isnull().sum()
```

```
TV              0
Radio           0
Social Media    0
Influencer      0
Sales           0
dtype: int64
```

Statistical Summary

```
df.describe()
```

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

Check Categorical Variables

```
df['TV'].value_counts()

df['Influencer'].value_counts()
```

```
Influencer
Nano     157
Micro    151
Mega     137
Macro    127
Name: count, dtype: int64
```

Data Preprocessing

Since TV and Influencer are categorical variables, they must be converted into numerical format.

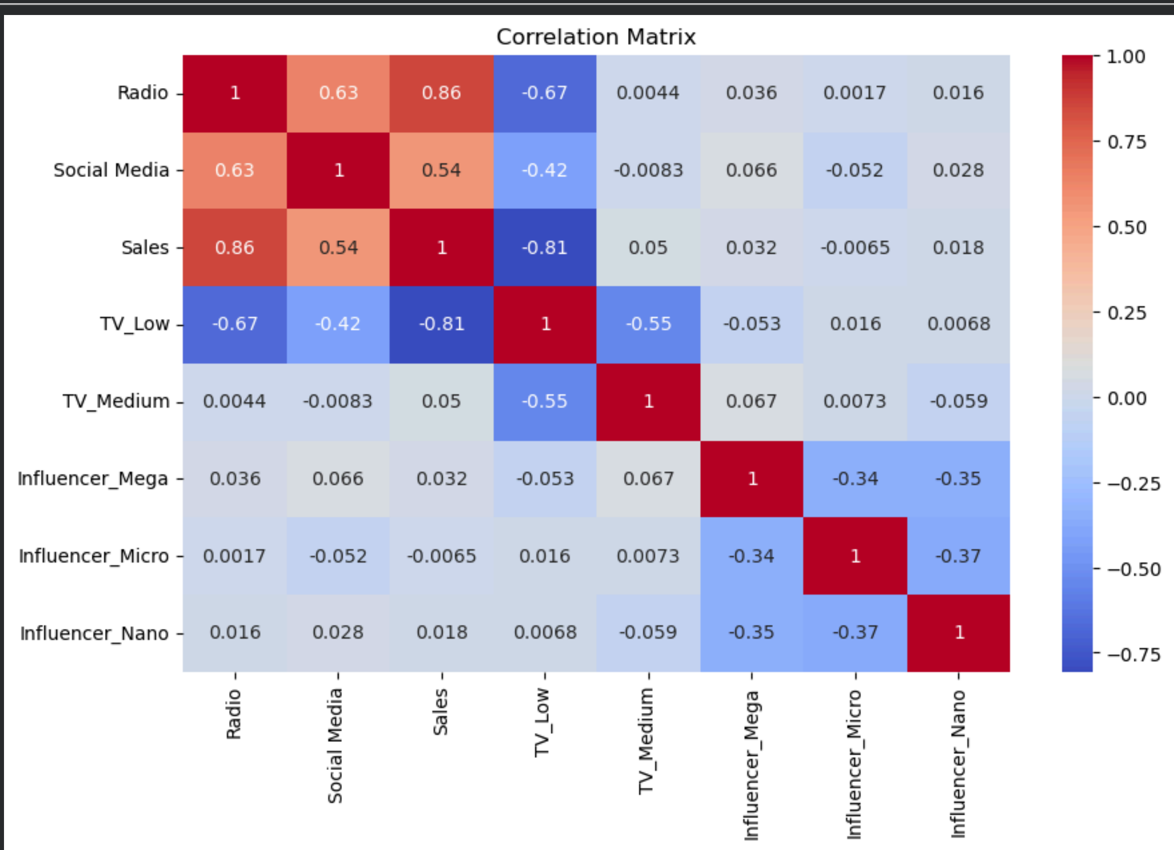
```
df_encoded = pd.get_dummies(
    df,
    columns=['TV', 'Influencer'],
    drop_first=True
)

df_encoded.head()
```

	Radio	Social Media	Sales	TV_Low	TV_Medium	Influencer_Mega	Influencer_Micro	Influencer_Nano
0	3.518070	2.293790	55.261284	True	False	False	True	False
1	7.756876	2.572287	67.574904	True	False	True	False	False
2	20.348988	1.227180	272.250108	False	False	False	True	False
3	20.108487	2.728374	195.102176	False	True	True	False	False
4	31.653200	7.776978	273.960377	False	False	False	False	True

Correlation Matrix

A correlation matrix helps identify relationships among predictors.

```
plt.figure(figsize=(10,6))

sns.heatmap(
    df_encoded.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Matrix')
plt.show()
```



Interpretation

* Strong correlations between predictors may indicate multicollinearity.

* Very high correlations (>0.8) require further investigation.

Check Multicollinearity Using VIF

Variance Inflation Factor (VIF) measures how much a variable is explained by other predictors.

Prepare Features

```
X = df_encoded.drop('Sales', axis=1)

X = sm.add_constant(X)
```

Calculate VIF

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
import pandas as pd
import statsmodels.api as sm

# Features only
X = df_encoded.drop('Sales', axis=1)

# Convert all columns to numeric
X = X.astype(float)

# Add constant
X = sm.add_constant(X)

# Calculate VIF
vif_data = pd.DataFrame()

vif_data["Feature"] = X.columns

vif_data["VIF"] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]

print(vif_data)
```

```
            Feature        VIF
0             const  31.649565
1             Radio   3.479641
2      Social Media   1.669098
3            TV_Low   4.076384
4         TV_Medium   2.233350
5   Influencer_Mega   1.593889
6  Influencer_Micro   1.618434
7   Influencer_Nano   1.627430
```

VIF Interpretation

VIF	Meaning
1-5	Acceptable
5-10	Moderate concern
>10	Serious multicollinearity

Results

Variable	VIF
Radio	3.48
Social Media	1.67
TV_Low	4.08
TV_Medium	2.23
Influencer Variables	~1.6

Conclusion: No severe multicollinearity exists in the dataset.

Build Multiple Linear Regression Model

Define Features and Target

```
X = df_encoded.drop('Sales', axis=1)

y = df_encoded['Sales']

X = sm.add_constant(X)
```

Train Model

```
# Features
X = df_encoded.drop('Sales', axis=1)

# Target
y = df_encoded['Sales']

# Convert everything to numeric
X = X.astype(float)
y = y.astype(float)

# Add intercept
X = sm.add_constant(X)

# Fit model
model = sm.OLS(y, X).fit()

print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.904
Model:                            OLS   Adj. R-squared:                  0.903
Method:                 Least Squares   F-statistic:                     760.4
Date:                Thu, 11 Jun 2026   Prob (F-statistic):         1.82e-282
Time:                        10:42:44   Log-Likelihood:                -2713.4
No. Observations:                 572   AIC:                             5443.
Df Residuals:                     564   BIC:                             5478.
Df Model:                           7
Covariance Type:            nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
const              217.4784      6.584     33.031      0.000     204.546     230.411
Radio                2.9735      0.235     12.644      0.000       2.512       3.435
Social Media        -0.1391      0.676     -0.206      0.837      -1.467       1.189
TV_Low            -154.5736      4.949    -31.231      0.000    -164.295    -144.852
TV_Medium          -75.5947      3.647    -20.726      0.000     -82.759     -68.431
Influencer_Mega      2.4948      3.462      0.721      0.471      -4.305       9.295
Influencer_Micro     2.9391      3.378      0.870      0.385      -3.695       9.574
Influencer_Nano      0.8015      3.346      0.240      0.811      -5.770       7.373
==============================================================================
Omnibus:                       58.711   Durbin-Watson:                   1.874
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               17.802
Skew:                           0.057   Prob(JB):                     0.000136
Kurtosis:                       2.143   Cond. No.                         147.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Model Evaluation

Adjusted R-Squared

```
print("Adjusted R-Squared:", model.rsquared_adj)
```

```
Adjusted R-Squared: 0.9030011006551627
```

Result

```
Adjusted R² = 0.903
```

Interpretation

The model explains approximately **90.3% of the variation in Sales**, indicating excellent predictive performance

Predictor Significance (P-values)

```
model.pvalues
```

```
const               5.906914e-134
Radio                1.943682e-32
Social Media         8.371166e-01
TV_Low              4.536854e-125
```

```
TV_Medium           2.277532e-71
Influencer_Mega     4.714413e-01
Influencer_Micro    3.845913e-01
Influencer_Nano     8.107454e-01
dtype: float64
```

Significant Variables

Variable	P-value
Radio	< 0.05
TV_Low	< 0.05
TV_Medium	< 0.05

Not Significant

Variable	P-value
Social Media	> 0.05
Influencer Categories	> 0.05

Interpretation

* Radio spending significantly impacts sales.

* TV advertising level significantly impacts sales.

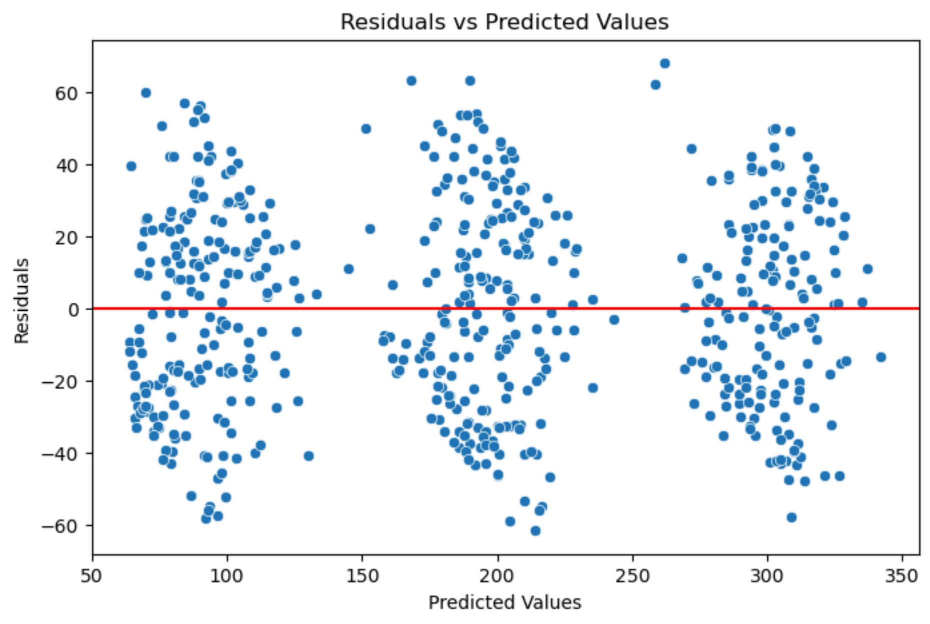
* Social Media spending is not statistically significant.

* Influencer category is not statistically significant.

Residual Diagnostics

A. Linearity Check

```
predictions = model.predict(X)

residuals = y - predictions

plt.figure(figsize=(8,5))

sns.scatterplot(
    x=predictions,
    y=residuals
)

plt.axhline(0, color='red')

plt.xlabel('Predicted Values')
plt.ylabel('Residuals')

plt.title('Residuals vs Predicted Values')
plt.show()
```

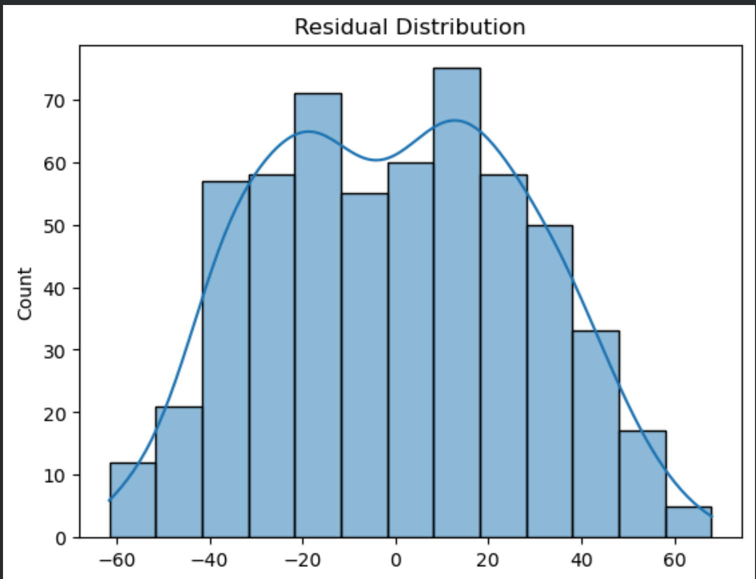


Expected Outcome

Residuals should appear randomly scattered around zero.

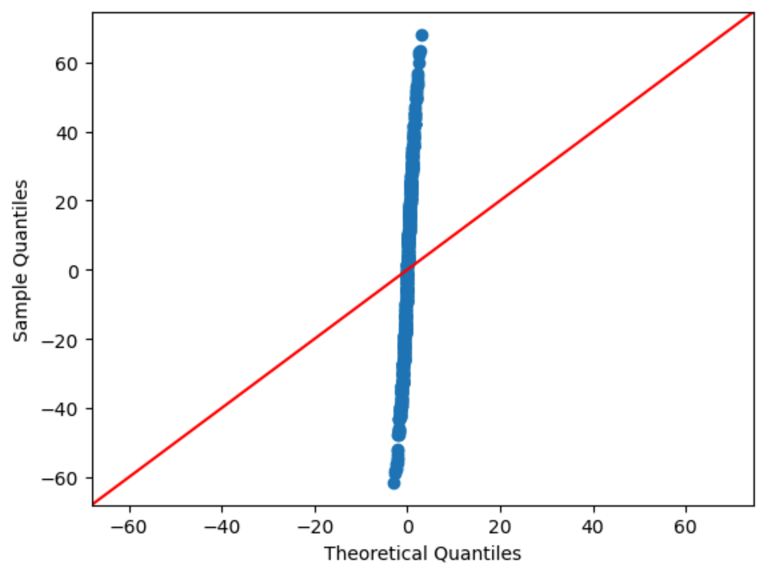
B. Normality Check

```
sns.histplot(
    residuals,
    kde=True
)

plt.title('Residual Distribution')
plt.show()
```



Q-Q Plot

```
sm.qqplot(
    residuals,
    line='45'
)

plt.show()
```

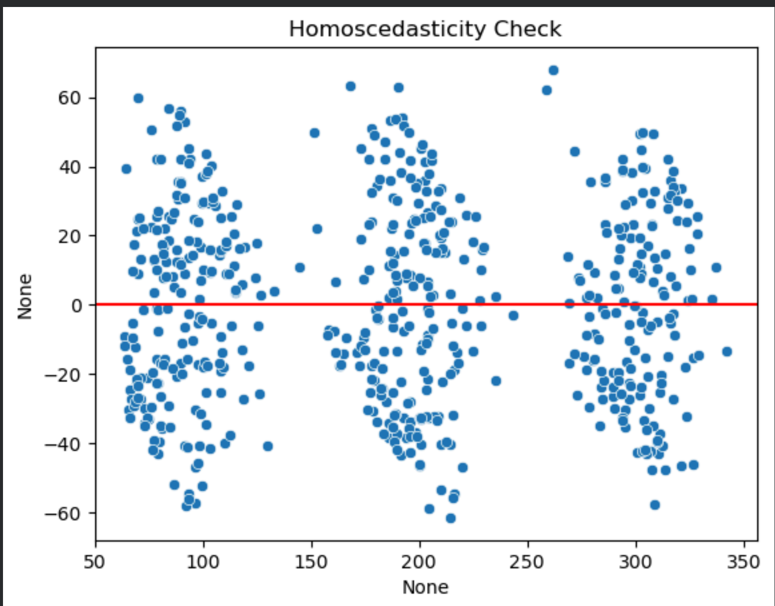


Expected Outcome

Points should follow the diagonal line closely.

C. Homoscedasticity Check

```
sns.scatterplot(
    x=predictions,
    y=residuals
)

plt.axhline(0, color='red')

plt.title('Homoscedasticity Check')
plt.show()
```



Expected Outcome

Residual spread should remain fairly constant across predictions.

Coefficient Interpretation

Radio

Coefficient ≈ **2.97**

Interpretation:

> Holding TV and Social Media spending constant, every additional 1-unit increase in Radio advertising spend is associated with approximately 2.97 additional units of Sales.

TV Categories

TV spending level has a substantial effect on Sales.

Compared to the baseline TV category:

* Low TV spend reduces expected Sales significantly.

* Medium TV spend also reduces Sales, though less severely.

* High TV spend generates the strongest sales performance.

Social Media

Coefficient ≈ **-0.14**

Interpretation:

> After accounting for TV and Radio advertising, Social Media spending shows no statistically significant effect on Sales.

Business Recommendations

Key Findings

1. Prioritize TV Advertising

TV spending shows the strongest relationship with Sales.

2. Continue Investing in Radio

Radio advertising has a statistically significant positive effect on Sales.

3. Reevaluate Social Media Spending

Social Media spend does not significantly improve Sales within this dataset.

4. Reassess Influencer Campaigns

Influencer categories do not significantly impact Sales and may not provide strong ROI.